Hye-Yoon Baek

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Research Interests

- Graph ML/DL
- Information Extraction
- Neuro-Symbolic LLM
- Logical Consistency of LLMs

Education

Konkuk University, Seoul, Republic of Korea *Mar. 2025 - Present*
Integrated M.S./Ph.D. in Computer Science and Engineering, GPA: 4.15/4.5
- Advisor: Prof. Byungkook Oh

Konkuk University, Seoul, Republic of Korea *Mar. 2019 - Feb. 2025*
B.S. in English Language and Literature & Smart ICT Convergence, GPA: 3.88/4.5

Experience

Graph & Language Intelligence Lab, Konkuk University *July. 2024 - Feb. 2025*
Undergraduate Research Intern

Publications

[5] **Hye-Yoon Baek**, Sangjun Ji, Jimyeung Seo, Hae-Yoon Koo, Xiongnan Jin, Byungkook Oh, "**Internalizing Negation-Gated Logical Rules into LLMs for Document-Level Relation Extraction**". In Findings of EMNLP 2026

[4] Jinho Choi, **Hye-Yoon Baek**, Sang-Hyun Park, Sangjune Kim, Hyeongjun Yang, Byungkook Oh, "**Causality-Guided Spurious Evidence Pruning with GFlowNet for Faithful Graph Retrieval**." In CIKM 2026

[3] Dongcheon Lee, Ji-Yeon Park, **Hye-Yoon Baek**, Jimyeung Seo, Seyeong Kim, Byungkook Oh, "**Diversifying Differentiable Graph Retrieval with Topic-Adaptive Multi-Intent Learning**." In WWW 2026

[2] Jimyeung Seo, Eun-Yeong Jo, **Hye-Yoon Baek**, Dongcheon Lee, Xiongnan Jin, Byungkook Oh, "**Image-Guided Debiasing Distillation with Preference Alignment Across Multi-News Histories**." In WSDM 2026

[1] **Hye-Yoon Baek**, Jinho Choi, Jimyeung Seo, Xiongnan Jin, Dongcheon Lee, Byungkook Oh, "**Relation-Faceted Graph Pooling with LLM Guidance for Dynamic Span-Aware Information Extraction**." In CIKM 2025

Patents

[2] **Hye-Yoon Baek**, Hae-Yoon Koo and Byungkook Oh, "**Method and Apparatus for Determining a Relation between Entity Pairs in a Document Using a Language Model**.",
KR Patent No.10-2026-0177145, filed September 16, 2026

[1] **Hye-Yoon Baek** and Byungkook Oh, "**Method and Apparatus for Dynamic Span-Aware Information Extraction Using Relation-Faceted Graph Pooling and LLM Guidance**.",
KR Patent No.10-2025-0162443, filed October 31, 2025

Research Projects

제조 도메인 지식그래프 기반 멀티턴 질의응답 기술 개발 *May 2026 – Dec. 2026*
LG전자 생산기술원 · 산업체 위탁
Project Member

멀티모달 생성모델의 신뢰성 및 제어가능성 향상을 위한 그래프 구조 기반의 상호작용 *Sep. 2024 – Aug. 2028*
한국연구재단 · 중견연구(유형1) · 개인연구
Project Manager

Language

- TOEIC 965 (26.03.16 exp)
- New TEPS 421 (Level 2+) (26.03.29 exp)